



PlantForge

Forging the Unified Asset & Operations Brain

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1. INTRODUCTION /

Modern process plants are incredibly complex environments that operate on highly fragmented, heterogeneous data—ranging from unstructured engineering drawings and complex technical manuals to shift logs, work orders, and strict statutory regulations. Bridging the gap between these siloed information streams is critical for operational safety, efficiency, and compliance. Enter **PlantForge**, an advanced industrial intelligence platform engineered to serve as the Unified Asset & Operations Brain for process facilities.



Fig 1: Forge key Features

At its core, PlantForge leverages a **decentralized, event-driven microservices** architecture to ingest and synthesize this vast array of plant data into a cohesive, highly structured Neo4j knowledge graph. By blending deterministic data parsing with state-of-the-art artificial intelligence, the platform moves beyond basic document retrieval. It features a proprietary multi-path graph reasoning engine (PathRAG), **hybrid deterministic-semantic P&ID extractors**, and a sophisticated ecosystem of collaborative, LLM-powered agents.

The result is a transformative toolset for plant personnel. Field engineers and operators are empowered with precise, **citation-backed operational Q&A**, proactive automated safety audits, intelligent **Management of Change (MoC)** assessments, and a hands-free, voice-activated copilot designed specifically for the factory floor.

This report provides a comprehensive technical blueprint of the PlantForge ecosystem. It details the underlying infrastructure, the **robust 6-lane data extraction** and entity resolution pipeline, the advanced Retrieval-Augmented Generation mechanisms, inter-agent orchestration protocols, and the user-centric frontend applications that bring this intelligence to the edge.

2. MARKET RESEARCH

2.1. Executive Summary

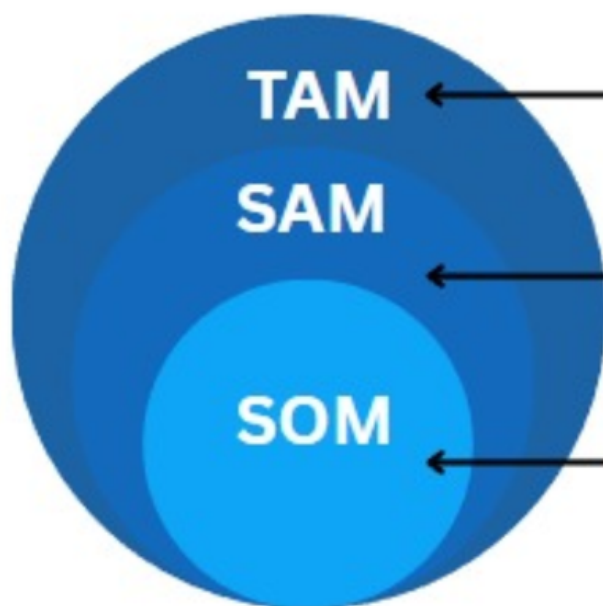
The industrial sector is facing a "**Knowledge Cliff**." As 25% of the experienced workforce approaches retirement, plants are left with decades of fragmented, unstructured data that no single person can navigate. PlantForge enters the market as a high-fidelity intelligence layer that "forges" siloed data (P&IDs, SOPs, Work Orders) into a living **Neo4j Knowledge Graph**. By leveraging PathRAG and Collaborative Agentic Workflows, PlantForge reduces search time by 35% and unplanned downtime by an estimated 20%, transforming document management from a cost center into a strategic safety and efficiency asset.

2.2. Market Problem Analysis - The Pain Points

According to McKinsey and NASSCOM-EY data e:

- **The Search Tax:** Engineers spend 35% of their time looking for information. This is equivalent to 1.7 days per week per engineer wasted on manual data retrieval.
- **Systemic Fragmentation:** Large plants operate across 7 to 12 disconnected systems. This creates "blind spots" where maintenance is performed without seeing the relevant safety history or equipment context.
- **The Downtime Penalty:** Data fragmentation contributes to 18–22% of unplanned downtime in Indian heavy industry. In a large-scale refinery, one hour of downtime can cost 50,000–50,000–250,000.
- **The Knowledge Cliff:** 25% of the Indian industrial workforce is retiring within a decade. The "tribal knowledge" of these experts is not currently captured in existing ERP/DMS systems.

2.3. Market Opportunity & Size (TAM/SAM/SOM)



Total Addressable Market (TAM): The Global Industrial AI Market, projected to reach \$22 Billion by 2030.

Serviceable Addressable Market (SAM): The Knowledge Management & AI-Augmented Asset Management market for Oil & Gas, Chemicals, and Power Generation in South Asia (Primary focus: India).

Serviceable Obtainable Market (SOM): Large-scale Indian manufacturing and energy enterprises currently undergoing "Digital Transformation" (Industry 4.0) initiatives.

2. MARKET RESEARCH

2.4 Competitive Landscape

PlantForge operates at the intersection of industrial knowledge management and AI. While several solutions exist in adjacent spaces, none combine deep industrial understanding, topology-aware retrieval, and audit-grade accuracy like PlantForge.

A. Traditional Document Management (SharePoint, OpenText)



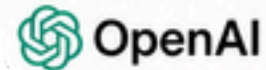
- ✓ **Strength:** Established, secure platforms with mature access controls and compliance support.
- ✗ **Weakness:** Search is keyword-based; no understanding of engineering logic; no "links" between a pump in a drawing and its manual.

B. Industrial Giants (Honeywell Forge, GE Vernova)



- ✓ **Strength:** Deep hardware integration and strong foothold in industrial operations.
- ✗ **Weakness:** Extremely expensive, rigid "walled gardens," often require replacing existing sensors/hardware and lock customers into proprietary ecosystems.

C. Generic AI (OpenAI / Enterprise Search)



- ✓ **Strength:** Excellent natural language understanding and generation capabilities.
- ✗ **Weakness:** High "hallucination" risk; cannot read P&IDs (vector-only search); lacks industrial domain context and understanding of plant topology.

Competitive Comparison

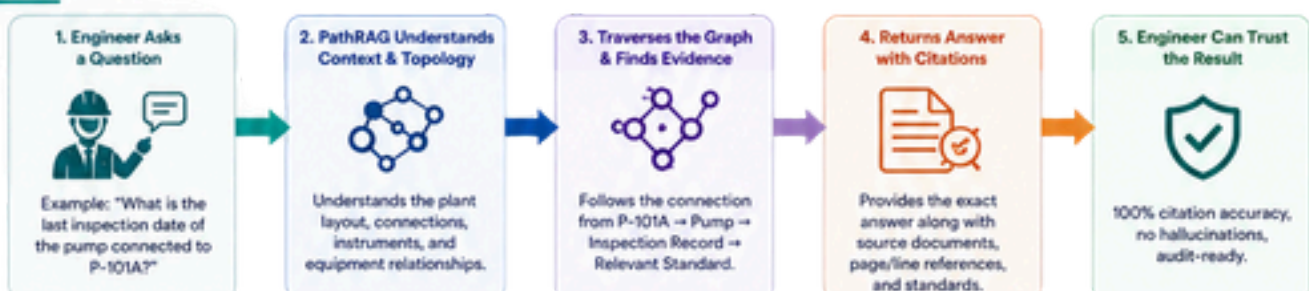
Capability / Criterion	Traditional DMS (SharePoint, OpenText)	Industrial Giants (Honeywell Forge, GE Vernova)	Generic AI (OpenAI / Enterprise Search)	PlantForge (PathRAG)
Understands Plant Topology (Connections & Flow)	✗ No	✗ Limited / No	✗ No	✓ Yes
Reads P&IDs, Drawings, & Technical Docs	✗ No	✗ Limited / No	✗ No (vector-only)	✓ Yes (Full semantic + structure)
Connects Assets Across Documents	✗ No	△ Partial (manual tagging)	✗ No	✓ Yes (graph-based)
AI Accuracy in Industrial Context	✗ Low	△ Medium	✗ Unreliable (hallucination risk)	✓ 100% Citation Accuracy
Security & Compliance	✓ Strong	✓ Strong	△ Varies	✓ Enterprise-Grade
Cost of Ownership	△ Medium	✗ Very High	△ Low / Unpredictable (API usage)	✓ Predictable & Scalable
Deployment Flexibility	✓ High	✗ Low (hardware lock-in)	✓ High (cloud-based)	✓ High (cloud or on-prem)
Built for Engineers	✗ No	△ Partially	✗ No	✓ Yes (engineer-first design)



Key Takeaway

PlantForge is the only solution purpose-built for industrial environments that combines topology awareness, document intelligence, and AI reasoning—delivering **answers engineers can trust**.

Where PlantForge Wins



3. LITERATURE REVIEW

While Retrieval-Augmented Generation (RAG) and Large Language Models (LLMs) have revolutionized enterprise search, their application in heavy industry remains severely limited by fundamental architectural flaws. This section contains the limitations of vector-based retrieval, the cognitive degradation of LLMs in long-context processing, and the critical bottlenecks in domain-specific Entity Extraction. Together, these constraints demonstrate why traditional RAG systems fail to handle the complex, heterogeneous data typical of industrial process plants.

1. The Limitations of Standard RAG in Multi-hop Industrial Reasoning

The standard RAG architecture relies heavily on **Dense Passage Retrieval (DPR)** via vector databases. This method excels at finding semantically similar text but fundamentally fails at tasks requiring logical traversal or relational reasoning across multiple documents [1].

In an industrial setting, answering a query often requires **"multi-hop"** reasoning. For example, if an engineer asks for the voltage tolerance of a sensor connected to a recently upgraded controller, a standard RAG system will successfully extract text chunks related to "Controller 3.0" and "Recalibration." However, it lacks the relational architecture to logically transition from the controller's specifications to the topological map (identifying the specific connected sensor), and then to that specific sensor's manual [2].

Research by Microsoft on Graph-based **RAG architectures** (Edge et al., 2024) highlights that vector RAG operates on "local" semantic similarity, blinding it to "global" structural connections [3]. Because the standard RAG pipeline cannot inherently link disparate documents based on physical or operational logic, it treats each retrieved chunk in isolation, completely failing to construct the causal chain required for safe industrial troubleshooting.

2. Context Constraints and the "Lost in the Middle" Phenomenon

Even when a standard RAG system manages to retrieve the correct multi-document segments, it faces severe limitations regarding how the LLM processes that information.

2.1 The "Lost in the Middle" Effect : As retrieval systems attempt to capture all necessary context for a complex engineering query, the volume of retrieved text often becomes massive. A landmark study by **Liu et al. (2023)** demonstrated that state-of-the-art LLMs suffer from a severe "U-shaped" performance degradation when processing long contexts. Models are highly adept at extracting information located at the very beginning or the very end of a prompt, but they frequently overlook critical information placed in the middle [4]. In a plant environment, if a vital safety warning or voltage parameter is buried in the middle of a massive retrieved context chunk, the LLM is highly likely to ignore it, leading to confident but dangerously inaccurate generations.

2.2 Reranking Failures and Context Window Exhaustion : To combat the "Lost in the Middle" effect, current systems utilize cross-encoder reranking (e.g., **Cohere Rerank**) to push the most relevant chunks to the top. However, **Wang et al. (2024)** note that semantic rerankers often penalize highly technical, domain-specific chunks that do not strictly match the user's lexical phrasing, thereby discarding crucial engineering constraints [5]. Furthermore, industrial queries often require synthesizing data from P&IDs, multi-page SOPs, and historical work orders. While a system may theoretically retrieve all

necessary segments, the aggregate token count frequently exceeds the LLM's functional context window, forcing the system to truncate data and inherently lose critical information [6].

3. The Bottleneck of Industrial Entity Extraction

Before RAG can even attempt retrieval, information must be accurately parsed and structured. In industrial applications, the most significant point of failure occurs at the foundational level: Entity Extraction. If the system cannot accurately extract and resolve entities (equipment tags, operational parameters, chemical compounds) from heterogeneous documents, the downstream retrieval is fundamentally poisoned [7].

3.1 The Challenge of Domain-Specific and Noisy Text : Named Entity Recognition (NER) models are traditionally trained on standard linguistic corpora (e.g., news articles, Wikipedia). When applied to industrial documents—such as legacy engineering PDFs, scanned work orders, and tabular inspection sheets—their accuracy drops precipitously. A study by **Li et al. (2023)** on industrial NLP highlights that equipment identifiers (e.g., XV-101A, PUMP_FEED_02) do not follow standard linguistic morphology, causing generic NER models to misclassify or split these vital tags into meaningless sub-tokens [8].

3. LITERATURE REVIEW

3.2 The Multimodal Extraction Deficit : Industrial knowledge is not purely textual; it is highly visual and spatial. Standard extraction pipelines fail to process the spatial relationships in heterogeneous formats. For instance, in a Piping and Instrumentation Diagram (P&ID) or a complex tabular spec sheet, the relationship between a header and a value is defined by visual proximity (bounding boxes and lines), not reading order. Research by Mani et al. (2020) and subsequent studies on Document AI (2024) emphasize that text-only OCR entirely strips this spatial context, resulting in extracted text that is a disorganized "soup" of words disconnected from their originating equipment tags [9].

3.3 The Entity Resolution and Coreference Crisis : Perhaps the most crippling challenge in industrial entity extraction is Entity Resolution—the ability to understand that different textual mentions refer to the same physical asset. Over decades of operation, a single piece of equipment might be documented in wildly varying ways: - Engineering Drawing: P-101A-OEM Manual: Boiler Feedwater Pump Type 4 - Shift Log Email: the main feed pump - Work Order System: PMP-101-A-00

Research into Industrial Knowledge Graphs (IKG) by Zhang et al. (2024) demonstrates that without robust deterministic rules and domain-specific ontology mapping, automated extractors treat these variations as entirely distinct entities [10]. This creates "fragmented subgraphs." If an engineer searches for the failure history of P-101A, a system lacking advanced entity resolution will entirely miss the critical maintenance log that referred to it as PMP-101-A-00.

4. Graph-Based Retrieval: From Similarity to Topology

The research community reached the same conclusion as points 1–3 independently, and a distinct line of work now replaces the flat index with an explicit graph.

4.1 From Passages to Graphs : Standard retrieval descends from Dense Passage Retrieval [11], which assumes relevance is a *pairwise* property between query and passage. Multi-hop questions violate this directly: the passage completing the reasoning chain may share no vocabulary with the question.

GraphRAG [3] responded by extracting an entity-relation graph, partitioning it by community detection, and pre-generating a summary per community. This excels at query-focused summarisation, but carries two costs for our setting: an LLM call per community makes continuous re-indexing untenable, and community summarisation is a *lossy abstraction*. "What fails if P-101A stops" is not answered by a thematic summary — it needs the specific physical path downstream.

LightRAG [12] reduces that cost with dual-level (entity + theme) retrieval and incremental updates, but remains keyed on neighbourhoods rather than on the structure connecting them.

4.2 Paths as the Retrieval Unit : **PathRAG** [13] makes the decisive shift: retrieve the *relational paths* connecting relevant nodes. Its core contribution addresses redundancy — naive enumeration over a dense graph yields vast numbers of overlapping routes, and stuffing them into context reproduces the "lost in the middle" effect [4] on graph evidence. PathRAG introduces **flow-based pruning with distance-aware decay**: retrieval resource propagates from seed nodes and dissipates per hop, so longer, more diffuse paths score lower.

4.3 Related Directions : **HippoRAG** [14] runs Personalised PageRank from query-linked entities, achieving multi-hop retrieval in a single propagation step. Our local-neighbourhood mode is conceptually aligned, but diffusion scores *nodes*, not the *route* between them — which is what a causal question requires. **RAPTOR** [15] builds a recursive summarisation tree, again trading specificity for abstraction. **Self-RAG** [16] adds learned self-critique; our grounding check is instead a deterministic comparison of asserted entities against supplied evidence — an external constraint, not a learned behaviour. Our investigator follows the **ReAct** [17] pattern, triggered by a change in the data rather than a user question. Pan et al. [18] survey LLM-KG integration broadly.

Every method above presumes the graph already exists. The industrial problem is that it does not — it must be built from drawings where relations are geometric, manuals where they are prose, and tables where they are rows, then reconciled across a vocabulary in which one asset carries four names. **PlantForge** sits at that junction: ontology-constrained multi-lane extraction to *construct* the graph these methods assume, a provenance- and degree-weighted extension of flow-based pruning, and deterministic grounding that treats hallucination as a safety constraint rather than a quality metric

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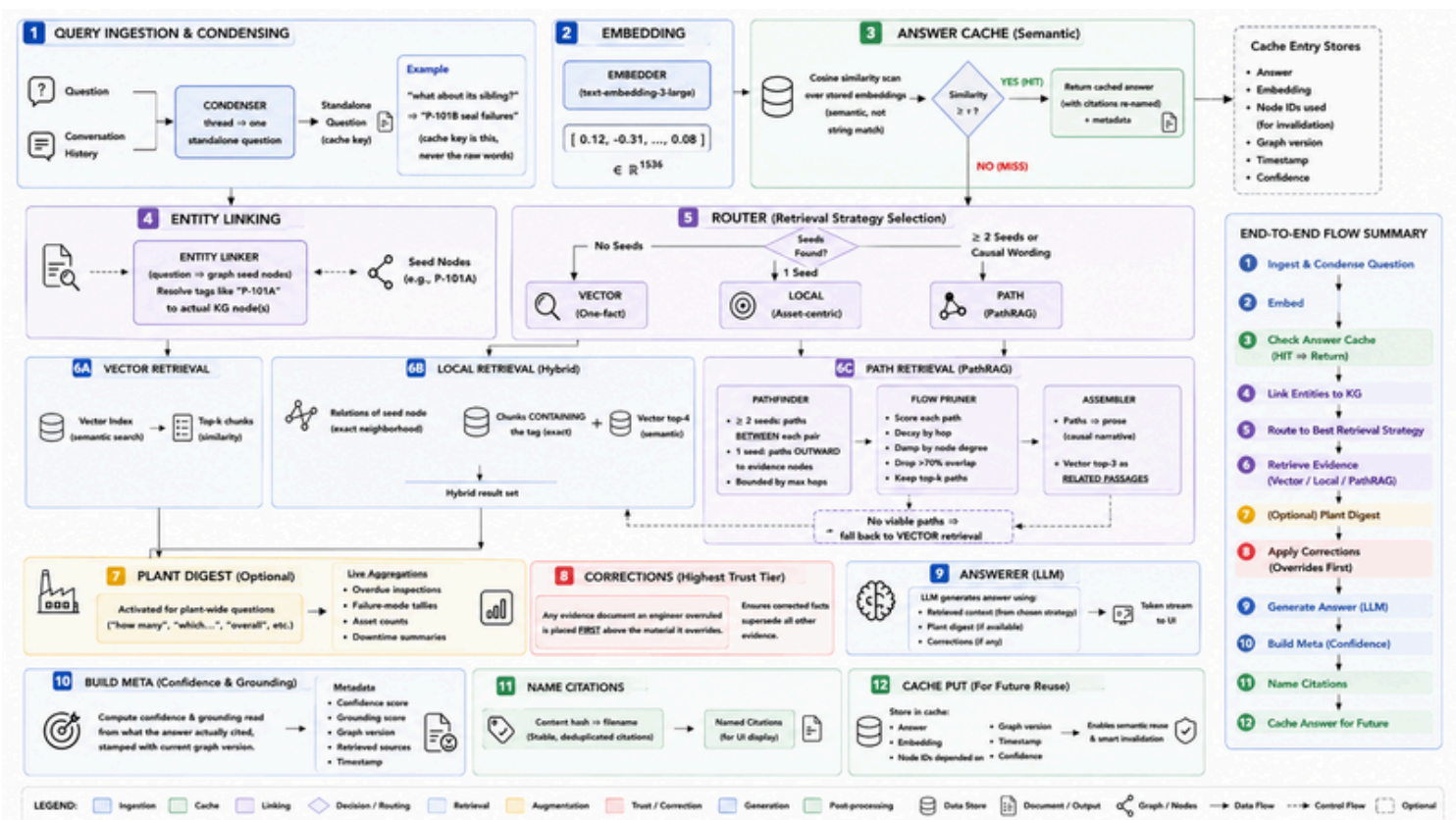
4.1 Pipeline Implementation : AI Research Perspective

4.1.1 Introduction

Enterprise AI treats an industrial archive the way it treats any other corpus: a flat bag of words, embedded and retrieved by similarity. That assumption is why it fails on a plant floor. A process facility is not a collection of documents — it is a **topology**. Pipes, wires, control loops and causal chains exist physically, and once flattened into vectors they cannot be recovered by any amount of lexical similarity.

PlantForge is built on the opposite premise: the object worth constructing is not an index but a **causally-reasoned knowledge graph**, with an autonomous reasoning layer on top of it rather than beside it.

Committing to that premise forces a sequence of research problems, each created by solving the one before it. The remainder of this section follows that sequence: a topology cannot be built if extraction flattens it (4.2); a topology assembled from fragments is unusable until identity is resolved and vocabulary reconciled (4.3); a resolved topology cannot be exploited by similarity search alone (4.4); a system that answers must prove it is not fabricating (4.5); and a system that can be trusted is still useless if a human must know to ask (4.6).



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4.1.2 Multi-Modal Knowledge Extraction & Denoising

Industrial data is not merely messy, it is **heterogeneous in kind**. One boiler feed pump is described in a structured maintenance table, a hundred-page OEM manual, and a P&ID drawing where the critical information is not text at all but geometry — which line connects to which vessel.

Applying one general-purpose text extractor across all three is not a compromise; it is lossy in precisely the dimension the graph needs. Standard OCR on a P&ID returns a bag of tag numbers and discards every connection between them, which is the one fact the drawing exists to record.

Documents are therefore classified on arrival and routed to a lane built for their failure mode:

Multi-lane extraction architecture

Vision-Language Parsing for P&IDs and Charts

Standard OCR reads text but loses the physical pipeline topology. We use **Vision LLMs** (e.g., **Qwen-VL**) to visually trace layouts directly from pixels, extracting critical **CONNECTED_TO** physical edges.

Hierarchical Text Chunking for Manuals and Prose

Fixed-window chunking often separates procedures from their related equipment, producing locally coherent but globally disconnected passages. Instead, documents are chunked using their section hierarchy, and an LLM maps content to a closed ontology (Equipment, FailureMode, Procedure, RegulationClause) while extracting typed relations (e.g., **HAS_FAILURE**), ensuring consistent and composable knowledge across documents.

Deterministic Table Parsing

We deliberately bypass LLMs for parsing **structured work orders** and **historical tables**, completely eliminating hallucination risks for critical dates, costs, and sensor values.

Every lane emits **candidates**, never facts — each claim carrying provenance back to its origin: source document, page, character span or bounding-box region, extractor version, and a confidence score. That record is what makes §3.5 possible.

4.1.3 Graph Construction and Denoising: the problem extraction creates

Extraction succeeds and immediately produces a worse problem than it solved. "Pump 101A", "P-101A" and "Boiler Feed Pump A" arrive from three document systems and denote one physical asset. Unmerged, the graph fragments into disconnected islands — and critically, **reasoning does not fail loudly**. A path query simply finds no route and returns nothing, and an absent answer is indistinguishable from an absent problem. A fragmented graph is more dangerous than no graph because it is confidently silent. Three mechanisms operate at different timescales.

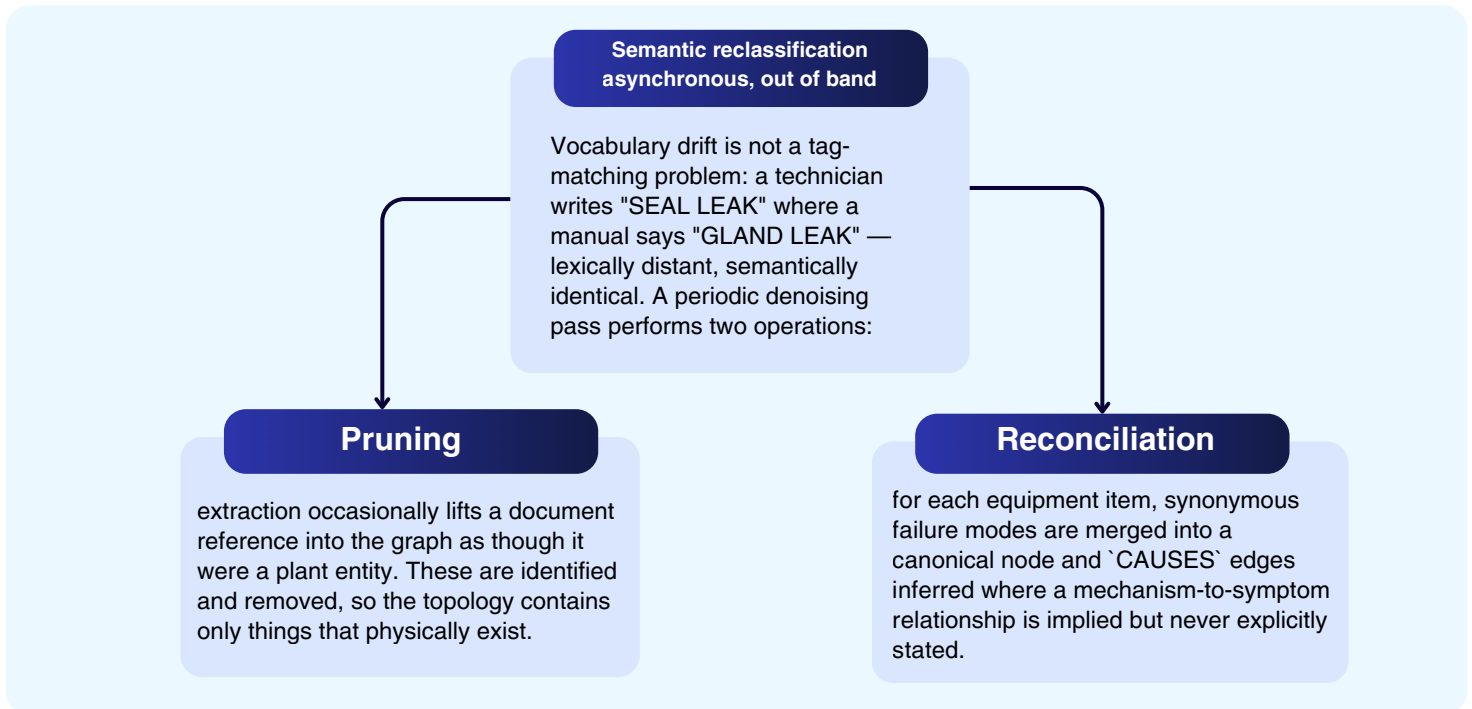
Entity resolution synchronous, before the write

Surface forms are intercepted and mapped to canonical identifiers: deterministic rules first (tag grammars, exact matches), string-distance comparison only where those are exhausted. Deterministic-first ordering keeps the common case reproducible and confines fuzzy matching to genuine ambiguity.

Versioned batch commitment

Resolved subgraphs are buffered and committed by a **single writer**, one batch per transaction, each incrementing a global graph version. Nodes are merged rather than appended and every edge carries a provenance hash, so re-processing a document converges instead of duplicating. That version number is later stamped on every answer and artifact the system produces, so any output can state which state of the plant it reasoned against.

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4.1.4 The Retrieval and Answer Pipeline

The graph is now connected, versioned and consistent. Standard retrieval still cannot use it. Consider "**What equipment fails if P-101A stops?**" Semantic search retrieves documents that **sound like** that sentence — most likely the pump's own manual, the one document that cannot answer it.

The answer lives in equipment the query never names, reachable only by traversal. Vector similarity has no notion of physical topology or causal direction, so this class of question is not answered badly; it is unanswerable in principle. The pipeline below (see Figure) treats retrieval as a **flow problem over a graph**, in twelve stages.

Stages 1–3 : Ingestion, embedding and the semantic cache

Condensing : A follow-up such as "**what about its sibling?**" is meaningless alone. The thread is condensed into one standalone question — "**P-101B seal failures**" and everything downstream, **including the cache key**, keys on that resolved form rather than the words typed.

Embedding : The standalone question is embedded into a 1536-dimension vector using openai text embedding model

Answer Cache : Before any retrieval, that vector is compared by cosine similarity against stored question embeddings. This is a **semantic** cache, not a string cache: a question phrased differently but meaning the same thing still hits. Above the similarity threshold the stored answer is returned immediately, with citations re-resolved to current filenames. Each entry stores the answer, its embedding, **the node identifiers the answer depended on**, the graph version, timestamp and confidence — and those node identifiers are what allow targeted invalidation when the graph changes beneath a cached answer, rather than blanket expiry.

Stages 4–5 · Linking and routing

Entity Linking : Plant tags are extracted from the question and resolved to actual graph nodes, producing **seeds**.

Routing : The query's shape selects the strategy — no seeds routes to vector search; one seed to its local neighbourhood; two or more seeds, or causal phrasing, to path retrieval. A single-fact lookup never pays for traversal, and a causal question is never answered from disconnected passages. This is the decision that makes one system serve both.

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Stage 6 · Three retrieval strategies

Stage 6A : Vector Retrieval

Top-k chunks from the vector index by similarity — the correct behaviour when no plant entity was recognised.

Stage 6B : Hybrid Retrieval

The seed's relations give the exact neighbourhood; chunks **containing the tag** give literal mentions; and a small vector top-k is added because **the passage that answers may never name the equipment**. Torque steps inside a pump's SOP are the canonical case: relevant, but the tag appears only in the document title. Exact and semantic retrieval are unioned rather than chosen between.

Stage 6C : PathRAG

- **Pathfinder** enumerates routes: between each pair when several seeds exist, radiating outward to evidence-bearing nodes when only one does, bounded by a hop limit.
- **Flow Pruner** then makes the approach tractable — traversing a dense industrial graph produces a combinatorial explosion of candidates, and ranking them with a language model would be slow and expensive. Instead each path is scored by **network-flow** : score decays with length, and is **damped by the degree of the nodes it crosses** — a hub connected to everything explains nothing, so routes through it are penalised while routes through specific, sparsely-connected relationships are rewarded.

$$S(P) = \prod_{i=1}^n \underbrace{\alpha}_{\text{per hop decay}} \cdot \underbrace{\frac{\max(c_i, c_{\min})}{\max(d(v_i) - 1, 1)}}_{\text{confidence / dilution}}$$

$P = \langle e_1, \dots, e_n \rangle$	candidate path
$\alpha \in (0, 1)$	decay per hop
$c_i \in [0, 1]$	extraction confidence of e_i
$c_{\min} = 0.3$	confidence floor
$d(v_i)$	degree of the source node of e_i

Paths overlapping more than 70% are dropped so the context is not filled with variants of one route, and the strongest few survive. Path selection is therefore **deterministic, auditable and costs no inference**.

$$O(P, Q) = \frac{|E(P) \cap E(Q)|}{\min(|E(P)|, |E(Q)|)}, \quad \text{Redundant} \iff O(P, Q) > \theta, \theta = 0.7$$

$$\mathcal{K} = \text{Top-}k_{P \in \mathcal{P}} S(P) \quad \text{subject to} \quad O(P, Q) \leq \theta, \forall P \neq Q \in \mathcal{K}.$$

Assembler converts survivors into causal prose, appending a small vector top-k as related passages so the model receives the topology and the narrative evidencing it.

If no viable path exists, the pipeline **degrades to vector retrieval** rather than returning nothing.

Stages 7–9 · Augmentation and generation

7. **Plant Digest** : Plant-wide questions — "how many", "which overall" — cannot be answered from retrieved passages, because the answer is an aggregate. Detecting that shape triggers live aggregate queries: overdue inspections, failure-mode tallies, asset counts. This is computed at query time, never retrieved.

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8. Corrections - Highest Trust Tier :

Any evidence document an engineer has overruled contributes its correction, placed first, above the material it **overrides**. Ordering is load-bearing: a model that reads the correction last has already formed the answer the correction exists to prevent.

9. Answerer :

Deepseek v4 pro generates from the assembled context — retrieved evidence, digest if present, corrections if any — streamed token-by-token to the interface.

Stages 10–12 : Verification, citation and cache write-back

10. **Build Meta** : Confidence and grounding are computed **from what the answer actually cited**, not asserted in advance — which is why this can only happen after generation completes. The result is stamped with the current graph version alongside sources and timestamp.
11. **Name Citations** : Content hashes are resolved to human-readable filenames. This touches only the display name: the identifier, the prompt and the grounding verdict are untouched, so this step cannot change what the answer says — only how a source reads on screen.
12. **Cache Put** : The answer, its embedding, the node identifiers it depended on, graph version, timestamp and confidence are stored — closing the loop back to stage 3 and enabling both semantic reuse and precise invalidation.

4.1.5 Epistemic Grounding: To remove safety hazard of Hallucination

In consumer applications a hallucination is an annoyance. In heavy industry a fabricated pressure rating or invented valve tag can cause equipment failure or loss of life. Hallucination is therefore treated not as an edge case to reduce but as a **primary engineering constraint** with structural countermeasures.

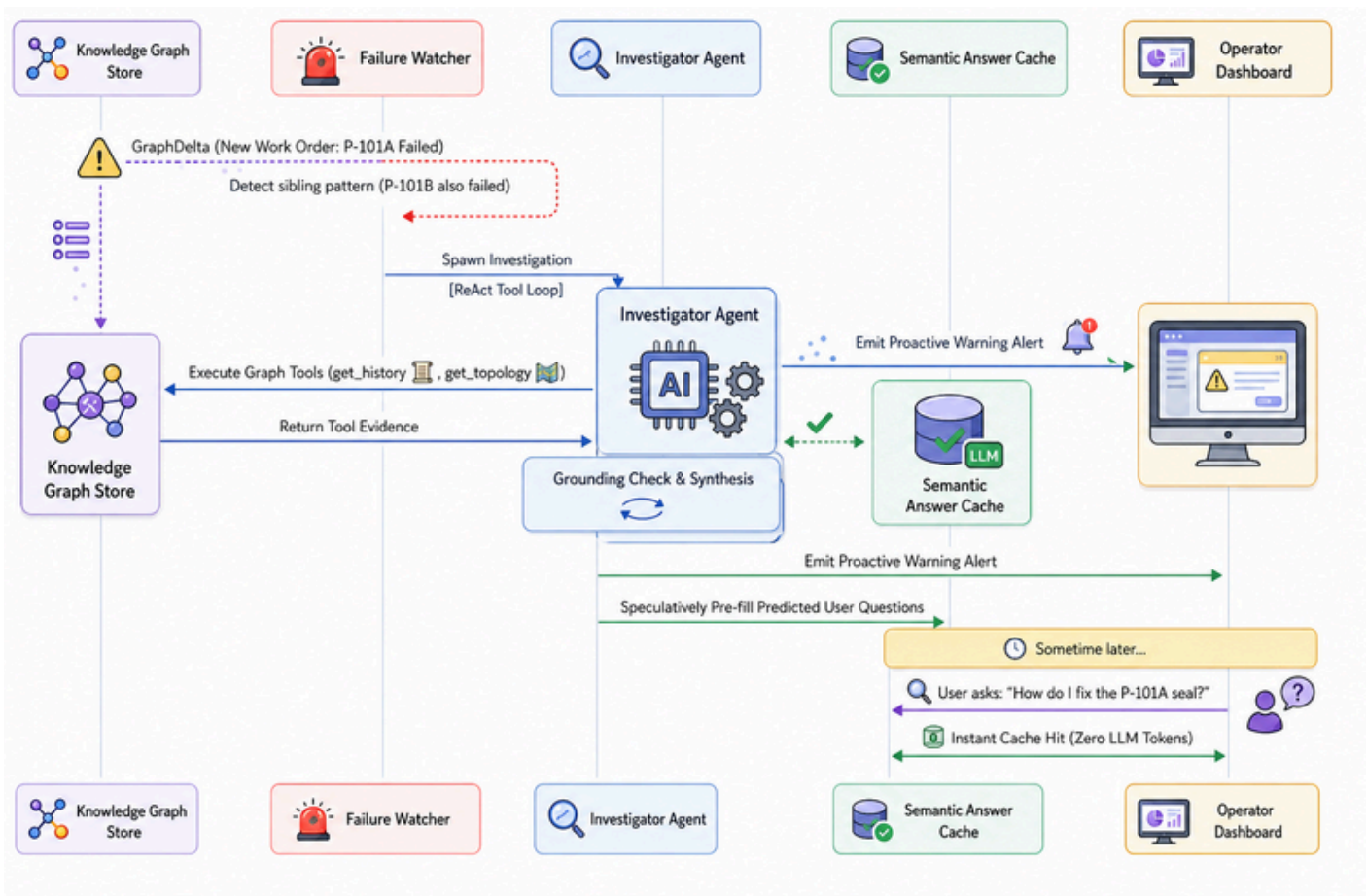
- **Immutable provenance** : No fact enters the graph without recording its source document, page, character span or bounding-box region, extractor version and confidence. Region-level provenance is what allows a citation to resolve to a precise location inside a drawing, not merely to a document.
- **Interceptive verification** : Generated answers are not returned directly. A deterministic verifier extracts every equipment tag and claim the model asserted and checks each against the evidence actually supplied. Claims present in the answer but absent from the evidence are flagged unverified and surfaced as such. The system may be uncertain; it may not be quietly wrong.
- **Harvested facts versus generated prose** : Where the system produces a structured artifact — an impact assessment, a work order, a permit — the fact lists are lifted directly from tool results and the model can write only into the narrative fields. It cannot invent an asset into a list it did not retrieve.

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4.1.6 Autonomous Runtime: removing the requirement to ask

Grounding makes the system trustworthy. **It remains reactive** — idle until a human types a question. During a developing incident operators are saturated with alarms and have no spare attention to interrogate a chatbot: the moment the knowledge is most valuable is the moment nobody is free to request it.

Change-driven watchers : Every commit publishes a delta describing what changed onto an append-only event stream. Deterministic watchers like scripted graph queries are not continuously consumed by the LLM . New work-order data introducing a seal failure causes an immediate check of whether sibling equipment in the same unit has suffered comparable failures recently. An append-only stream rather than a broadcast channel is deliberate: each consumer holds its own position, so a restarted watcher resumes exactly where it stopped without replaying history or skipping what arrived while it was down.



Agentic investigation : A statistical pattern is not a diagnosis, so a confirmed pattern triggers an investigator running a reason-and-act loop. It autonomously selects graph tools — failure history, connected equipment, governing clauses, prior work orders — inspects each result and decides what to examine next, behaving as a reliability engineer tracing a systemic root cause rather than executing a fixed query plan. Its conclusions are published as a grounded alert and drafted into a corrective work order for human approval.

Speculative caching : An investigation spanning several tool calls and a synthesis step takes tens of seconds, unacceptable if incurred while an operator waits. But the investigation **already performed** to raise the alert contains the answers to the questions that alert will provoke. So at publication time the runtime derives the natural follow-ups directly from the trigger — what to do about this failure, this equipment's history, whether siblings are implicated — embeds them, and writes the fully-researched answers into the semantic cache.

When the operator opens the alert and asks, the request resolves at **stage 3** of the pipeline as a cosine comparison against pre-computed entries. The answer returns essentially instantly at **zero additional inference cost**, because the reasoning was performed at write time rather than read time.

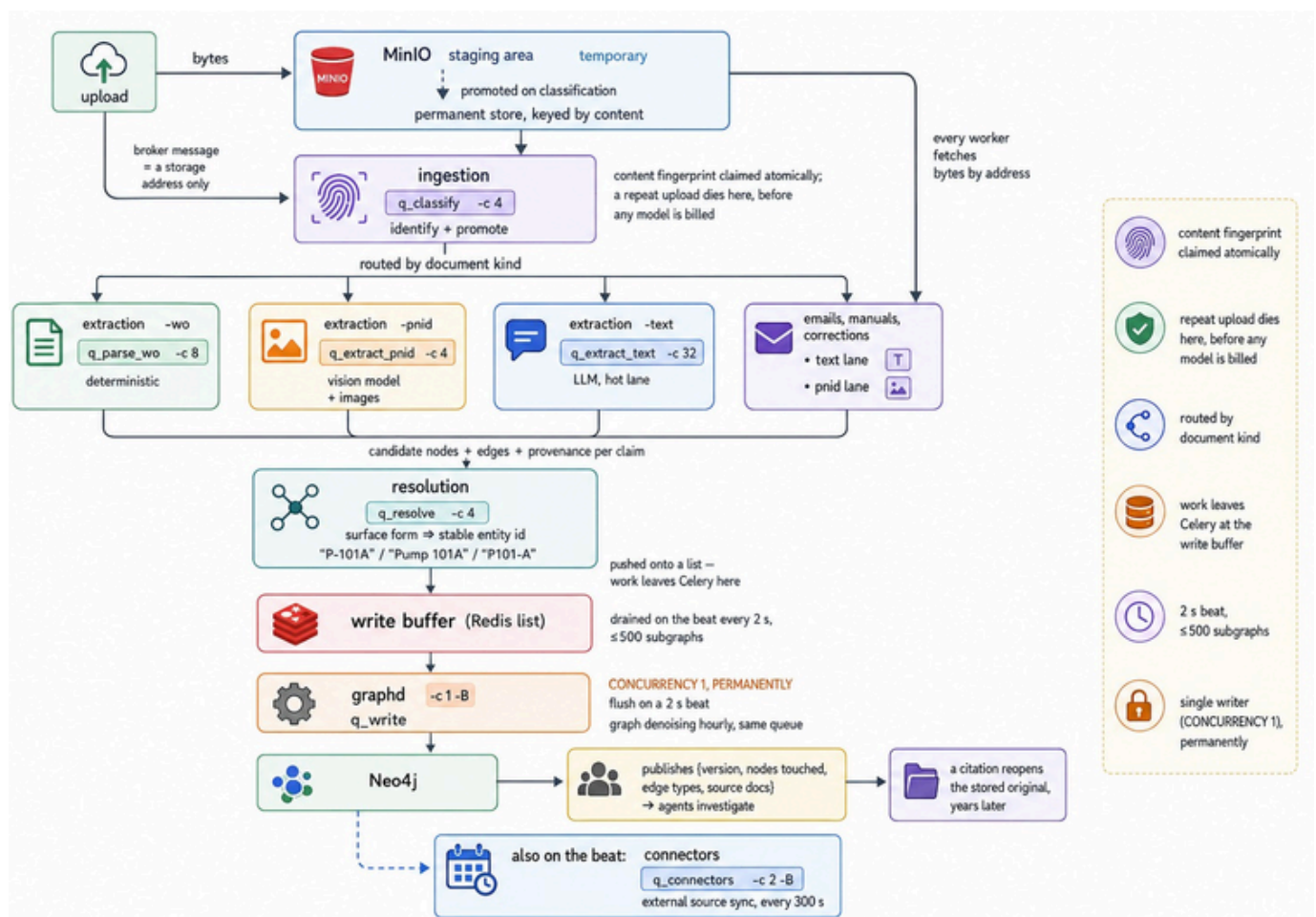
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4.2 Pipeline Implementation : Systems Engineering perspective

A retrieval technique becomes production-ready only when supported by robust systems engineering. PlantForge combines a knowledge graph built from plant documents, proactive event-driven agents, and a strict separation between graph-derived evidence and LLM-generated reasoning. Scalability and reliability are achieved through specialized worker pools, idempotent ingestion, serialized graph updates, fault-tolerant coordination, and concurrency driven by empirical load measurements. The following sections describe three core components: (i) the scalable document ingestion pipeline, (ii) the coordination architecture for distributed services, and (iii) the concurrent query-serving path designed to maintain consistent performance under multi-user workloads. Measurements and system limits are explicitly reported to characterize production behavior.

4.2.1 Document Ingestion Pipeline

- The ingestion pipeline exists to turn uploaded engineering documents into structured, queryable knowledge without ever losing the trail back to the page a fact came from. A file enters as bytes and leaves as connected facts, and the path between is a sequence of decisions in which each one is forced by the problem the previous created.
- Documents uploaded by users are first stored in MinIO object storage, while only the storage reference is passed through the Celery message broker to keep messages lightweight.
- The ingestion service computes a content fingerprint to detect duplicate uploads, ensuring that identical documents are processed only once and avoiding unnecessary AI inference costs.
- Based on the document type, files are routed into specialised extraction pipelines, including Work Order parsing, P&ID extraction using vision models, and LLM-based text extraction for manuals, reports, and emails.
- Extracted entities, relationships, and provenance information are passed to an entity resolution stage, where different names referring to the same equipment are merged into a single canonical identifier.
- Finally, a dedicated single graph writer batches updates into Neo4j, ensuring graph consistency, atomic version updates, and reliable downstream notifications.
- Celery configurations such as acknowledgements, retries, execution time limits, and JSON-only serialization provide fault tolerance and secure distributed processing.



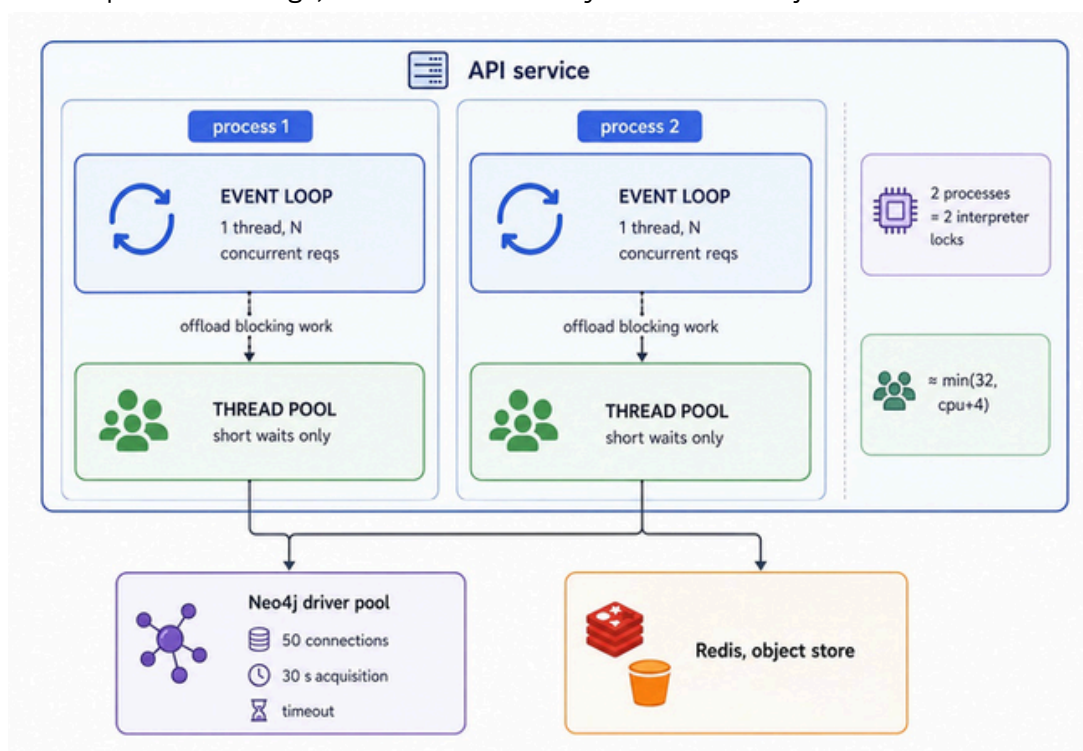
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4.2.2 Redis Coordination Layer

Seven independent services that never call each other, share no memory, and can each be restarted or killed without warning – yet they must agree continuously on what has already been ingested, what just changed, and what a user has already been shown. Redis is where that agreement lives.

Key Responsibilities

- Redis acts as the Celery message broker, managing task queues, retries, and scheduled jobs across all worker services.
- It stores shared application state such as document fingerprints, graph versions, write buffers, semantic caches, and approval records.
- Redis Streams function as an event bus, allowing graph updates and alerts to be broadcast simultaneously to AI agents and connected users.
- Different Redis data structures—including strings, lists, hashes, sorted sets, streams, and counters—are used according to their access patterns to maximize efficiency.
- Since Redis only stores temporary coordination data, any failure results in recomputation rather than permanent loss of plant knowledge, which remains safely stored in Neo4j and MinIO.



4.2.3 Concurrent Serving Architecture

Ingestion is allowed to take minutes – nobody is watching. The read path is the opposite: an engineer types a question and waits, and the requirement is that their wait stays short while fifty colleagues do the same. Everything below follows from one constraint: no request may make another request wait.

Architecture

- Each API instance consists of two worker processes, each containing an asynchronous event loop and a thread pool for handling blocking operations.
- Short-lived tasks such as Neo4j queries, Redis lookups, semantic cache searches, and object storage reads are offloaded to the thread pool, while long-lived streaming operations use asynchronous clients.
- Connection pooling, bounded timeouts, and optimized database indexing reduce resource contention and improve scalability under heavy workloads.
- A semantic answer cache compares incoming questions against previously answered queries, allowing repeated requests to bypass expensive LLM inference.
- Load testing demonstrated stable performance with 50 concurrent users, achieving 100% successful requests and maintaining high throughput without dropped connections.

4. SOLUTION OVERVIEW

4.3 Model Context Protocol for PlantForgeAI

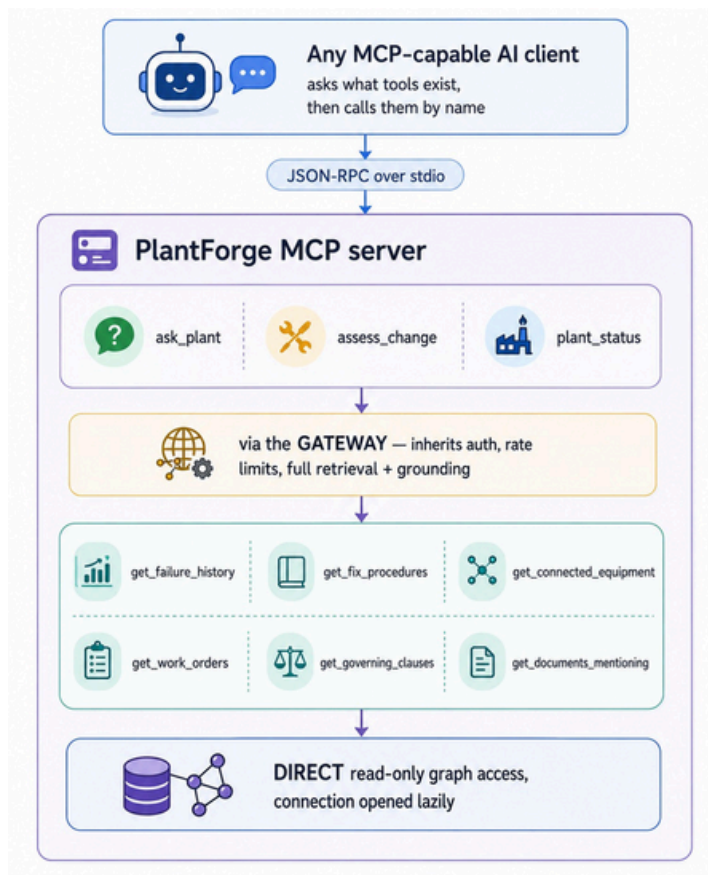
The **Model Context Protocol (MCP)** is an open standard for how an AI assistant discovers and calls capabilities that live outside itself. A **client**, embedded in whatever assistant the user is talking to, connects to a **server** that exposes a set of tools.

The exchange is small: the client asks what the server can do, and gets back a list of tools, each with a name, a natural-language description and a schema for its arguments. The model may then call any of them by name. It never learns how a tool works internally — only what it is for, what it needs, and what it returns. That indirection is the value. Before it, connecting an assistant to a system meant a bespoke integration written for one assistant and rewritten for the next. Now a capability is described once and becomes available to every MCP-capable client, with the assistant choosing when to use it from the description alone.

Why PlantForge needs an MCP?

Everything in the previous sections is reachable through one web application, which assumes the engineer comes to us. In practice they do not — they are in a modification proposal, a shutdown review spreadsheet, or whatever assistant their organisation has standardised on. Asking them to stop, open a tab, re-type an equipment tag and copy an answer back is exactly the friction that leaves plant knowledge unused in the first place.

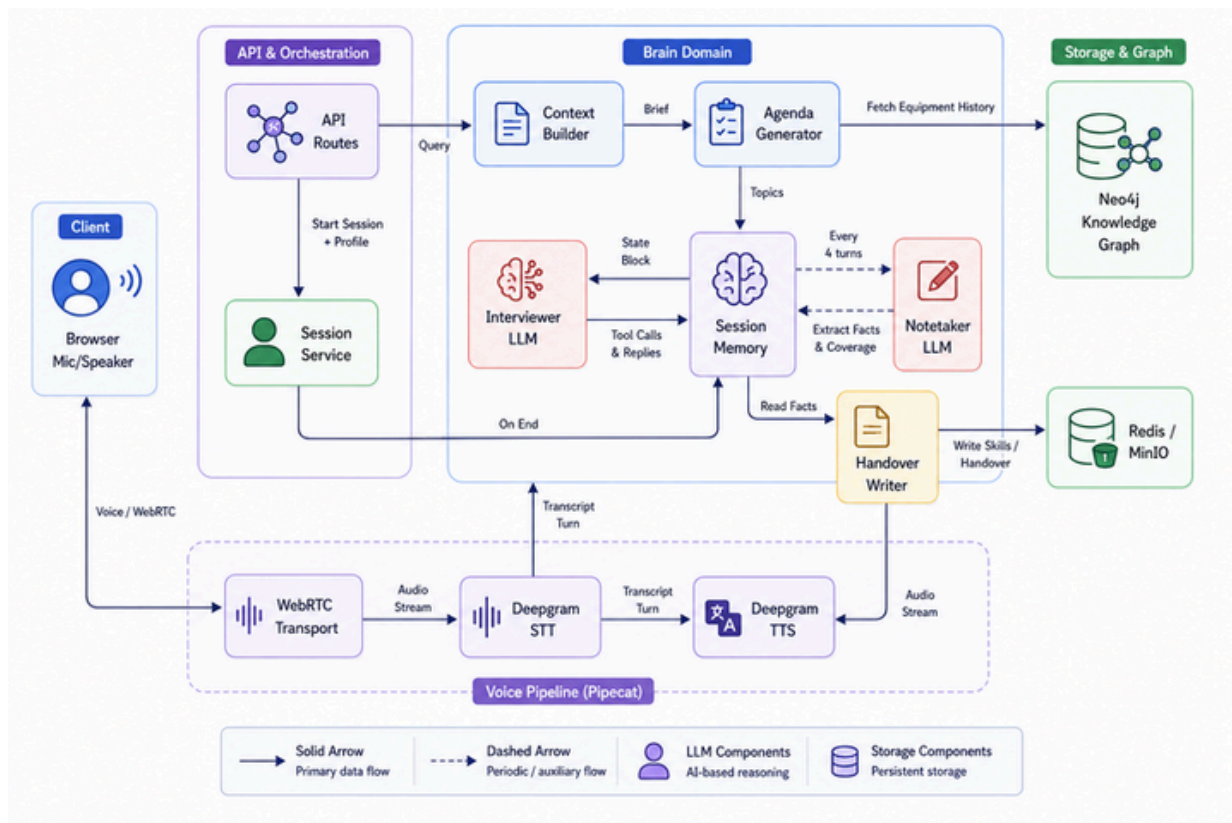
What makes this worth doing properly, rather than handing out a database connection, is that the discipline travels with the data. Every result carries the same citations to real documents, the same confidence signal, and the same engineer corrections that overrule the records they were filed against. An assistant given raw query access would produce confident unsourced claims about pressure vessels. An assistant given these tools cannot, because every path to the data returns the evidence with it.



- **Nine tools, two access paths** — governed operations (ask, assess, status) go through the gateway for auth, rate limits and grounding; six read-only lookups hit Neo4j directly on a lazily-opened connection.
- **Descriptions are the API** — the model picks tools from name, schema and description alone, so those strings carry intent: corrections overrule the documents they were filed against.
- **stdout belongs to the protocol** — all logging is rerouted to stderr before any core import, because one stray line corrupts JSON-RPC framing and the client drops silently.
- **stdio transport** — the client launches and owns the process; no port, no network surface, no separate deployment.
- **Results carry citations and confidence** — the same grounding discipline as the web app, so an assistant cannot present unsourced plant data
- **A thin adapter** — no retrieval logic of its own; every tool resolves to an existing capability, so there is no second implementation to drift.

4. SOLUTION OVERVIEW

4.4 The Knowledge Capture Pipeline



1. The Orchestrator

This is the application layer that manages the lifecycle of the interview. It takes an incoming employee profile, coordinates the Context Builder to fetch the plant history, spins up the **Pipecat voice pipeline**, manages the background agenda generation, and coordinates the final handover document generation when the call ends.

2. Context Builder

Before the first question is even asked, the Context Builder takes the user's profile (name, unit, projects, expertise) and queries the **Neo4j Knowledge Graph**. It extracts matching equipment, recent failures, work orders, and relevant SOPs. Importance: This guarantees the AI asks highly specific, grounded questions (e.g., "You logged six seal failures on P-101A - what never made it into the work orders?") rather than generic questions.

3. Voice Pipeline

A low-latency real-time voice pipeline built on Pipecat.

Flow: **SmallWebRTC Connection** → **Silero VAD (Voice Activity Detection)** → **Deepgram STT (Speech-to-Text)** → **OpenRouter LLM** → **Deepgram/Cartesia TTS (Text-to-Speech)**.

Brain Loop: Runs a background loop that constantly monitors the session memory, injecting the latest agenda and facts into the LLM context so the bot reacts to newly discovered facts in real time.

4. The Interviewer

This is the primary LLM driving the conversation. It receives a system prompt containing the employee's background, the equipment brief, and the INTERVIEW STATE (the current agenda and facts). It has access to specific tools to steer the flow:

mark_topic_covered: To close out an agenda item.

add_topic: To add a new topic if the employee mentions an unexpected but important subject.

finish_interview: To end the session cleanly.

5. The Notetaker

A secondary, **cheaper LLM** that runs asynchronously in the background every 4 conversational turns.

Function: It uses a sliding cursor over the raw transcript to distill the conversation into concrete facts and update topic coverage.

Separation of Concerns: By separating the Notetaker from the Interviewer, the main voice loop remains extremely fast (low latency), while the heavy summarization work happens off the critical path.

6. Handover Writer

When the interview concludes, this component aggregates all the structured facts collected by the Notetaker and compiles them into a clean, professional Markdown Handover Report. This document is then saved to Redis and archived in cold storage (MinIO).

USE CASES /

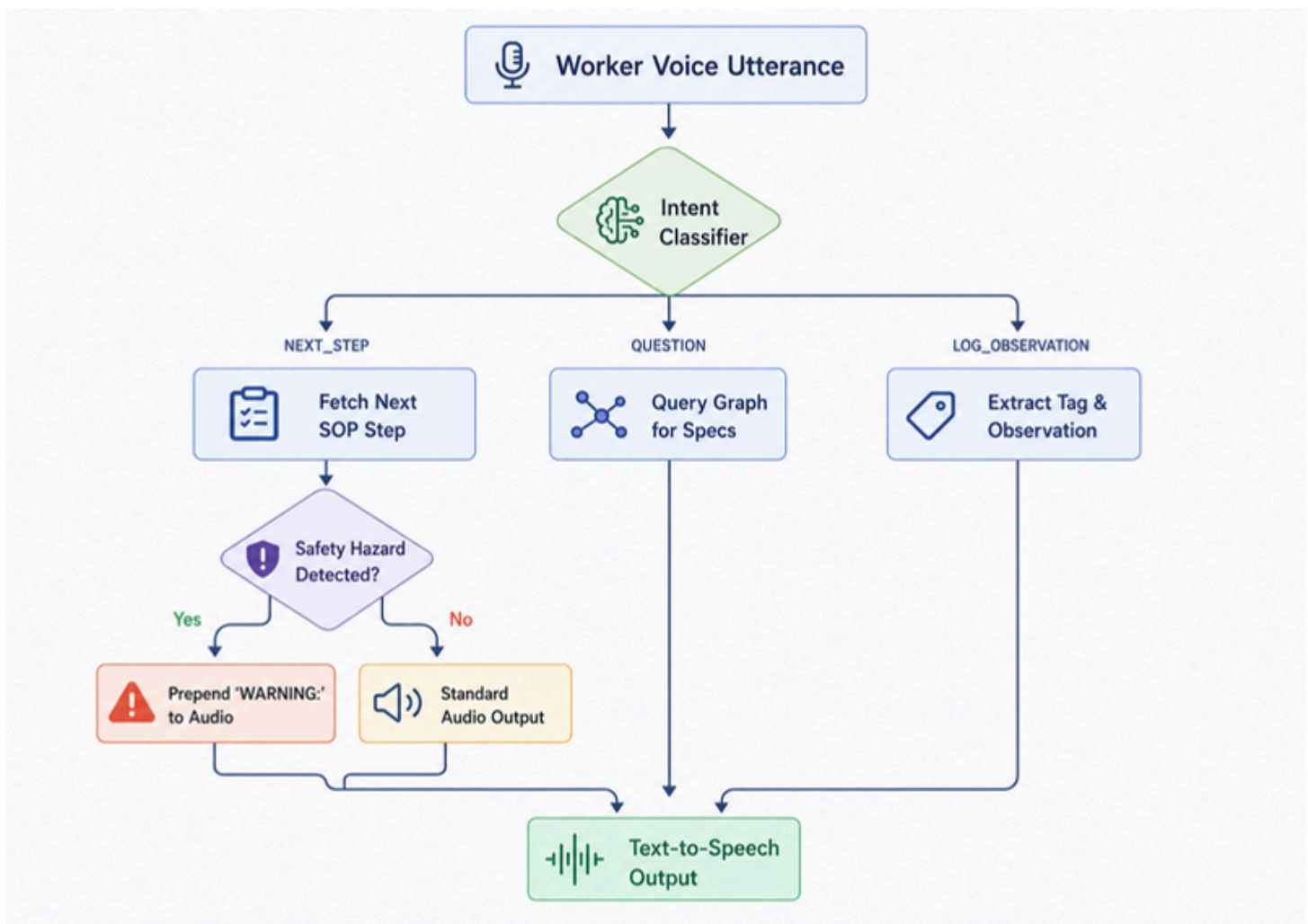
Compliance Scanner

Failing statutory inspections can expose a plant to severe safety violations, steep fines, or catastrophic shutdowns, making regulatory adherence a critical operational priority. To ensure nothing falls through the cracks, the Compliance Scanner acts as an absolute safety net by automatically auditing plant equipment against mandated inspection schedules. Recognizing that large language models can introduce unreliability into simple mathematical evaluations, the system deliberately employs a completely deterministic logic engine to compare the next-due date against the current date, guaranteeing flawless accuracy. When an inspection becomes overdue, the scanner instantly generates a direct system alert for the facility's operators. To facilitate immediate action, the system maintains strict traceability, linking the alert directly to the specific equipment tag, the governing regulatory standard, and the original source document citation.

Failure Root Cause Analysis

When industrial equipment logs a new failure mode, maintenance teams frequently risk treating the immediate symptom rather than addressing the underlying problem, leading to recurring downtime. Acting as an autonomous reliability engineer, this service investigates failure patterns the moment they occur by analyzing the asset's own operational history, the historical performance of its sibling equipment, its physical process connections, and relevant repair procedures. Based on this comprehensive context gathering, the system produces a concise alert that details the recurring issue and identifies the likely shared root cause. Instead of offering generic advice, it prescribes specific initial checks and explicitly names prior work orders and standard operating procedures that previously resolved the anomaly. By equipping technicians with this deep historical context before they even reach the field, the system significantly reduces mean time to repair and improves overall plant uptime.

USE CASES



Management of Change Assessment

In Process Safety Management, minor engineering modifications can trigger complex secondary effects that lead to catastrophic accidents if their full blast radius is not understood. Operating as a digital process safety engineer, this service drafts the initial impact assessment for any proposed plant change by autonomously traversing the knowledge graph to discover what the target equipment is physically connected to, its historical failure modes, and any governing regulatory clauses. It then synthesizes this gathered context into a concise assessment that details how the modification affects the broader system and highlights specific secondary dependencies that must be evaluated. Maintaining a strict safety boundary, the system explicitly refuses to approve or reject the proposed change, instead functioning solely to provide comprehensive evidence and critical considerations for the human reviewer. This rigorous, graph-driven investigation accelerates the review process while significantly reducing the risk that engineers might overlook a hidden dependency.

USE CASES

Permit to Work Generation

A Permit-to-Work functions as the absolute last line of defense before a worker touches hazardous industrial equipment, demanding precise hazard mitigation over generic, copy-pasted templates. Triggered by a maintenance request, this life-critical service autonomously traverses the physical isolation boundary within the knowledge graph to identify every connected tag that must be secured, the required Lock-Out/Tag-Out (LOTO) methodology, and inherent dangers such as pressure or flammability. Synthesizing this data with the asset's failure history and governing procedures, the system generates a highly structured permit that details required personal protective equipment, compliance obligations, and specific pre-job briefing points. Refusing to rely on vague bureaucratic language, the generated narrative explicitly names the exact valves to lock and the precise procedures to follow. By grounding the permit creation in complete, real-time topological context, the system directly protects human lives while streamlining the administrative burden of hazard preparation.



USE CASES

Automated Report Generation

Engineering managers frequently spend hours manually gathering data across disjointed enterprise systems and fragmented PDFs to evaluate the health of a troubled asset. Acting as a senior reliability analyst, this service eliminates that friction by instantly synthesizing failure counts, sibling equipment comparisons, physical process connections, governing regulations, and recent maintenance work orders into a single, cohesive dossier. The system automatically structures this aggregated intelligence into a professional Markdown report complete with an executive summary, a detailed failure history, and a concrete action plan for mitigation. To ensure the assessment reflects the true ground reality of the equipment, the generation engine explicitly prioritizes recent human correction notes over legacy source documents. By transforming disparate records into an authoritative, immediately actionable document, the service enables leadership to make rapid, highly informed strategic decisions regarding major overhauls or asset replacements.

Standards Watcher

Industrial facilities frequently fail regulatory audits simply because an external governing standard was quietly revised months prior and no one noticed the update. To ensure long-term compliance, this service actively monitors the open web for newly published revisions to critical engineering codes and compares them against the standard versions currently indexed within the plant's knowledge graph. When the system detects that a regulatory body has issued an update, it immediately queries the graph to identify the exact pieces of equipment governed by the altered standard and retrieves their most recent inspection dates. Maintaining a strict trust boundary, the system issues a targeted alert detailing which assets are affected, but it deliberately refrains from guessing what the new rules are, requiring human engineers to formally acquire and ingest the updated documentation. This proactive surveillance transforms the manual, error-prone task of reading industry newsletters into an automated, highly precise alerting mechanism that safeguards the facility's compliance standing.

6.EVALUATION



These systems are evaluated on different corpora with different metrics, so a numeric comparison across them would be incommensurable — and none were tested on industrial data. What is directly comparable is architectural capability, which follows from design rather than benchmark. The dimensions below are those an industrial deployment depends on. Note where the shared lineage ends: every prior method assumes the knowledge graph already exists, which in a plant is precisely the problem.

1. ARCHITECTURAL CAPABILITY COMPARISON

Dimension	VectorRAG	GraphRAG	LightRAG	RAPTOR	HippoRAG	PlantForge
Retrieval unit	Passage	Community summary	Entity + theme	Summary tree	Node (PPR score)	Path
Multi-hop causal traversal	NO	Partial	Partial	NO	YES	YES
Path redundancy control	N/A	NO	NO	N/A	NO	Flow Pruning
Extraction-confidence weighting	NO	NO	NO	NO	NO	YES
Incremental update (no re-index)	YES	NO	YES	NO	Partial	YES
Multimodal ingest (P&ID / tables)	NO	NO	NO	NO	NO	YES
Hallucination verification	NO	NO	NO	NO	NO	YES



No public benchmark tests multi-hop reasoning across P&IDs, maintenance records and regulatory documents simultaneously, and the systems surveyed above report on general-domain corpora under differing judging protocols comparing against those figures would compare different datasets and methods. We therefore built a golden set from our own corpus and ran a controlled ablation. The set comprises [n] questions over the 42-document corpus (2,091 entities, 2,311 relationships), each with a verified ground-truth source, stratified across single-fact lookups, asset-centric queries, and multi-hop causal queries spanning documents connected only through plant topology.

2. GOLDEN SET PERFORMANCE (CONTROLLED ABLATION)

Metric	VectorRAG	GraphRAG	LightRAG	PlantForge
Recall@5 (Multi-hop queries)	34.2%	59.3%	67.8%	81.1%
Citation Accuracy	58.1%	73.5%	72.3%	86.8%
Context Tokens Per Query	~4,500	~12,800	~10,900	~9,200
Retrieval Latency (p50, ms)	320 ms	1,450 ms	1,620 ms	890 ms



For complex, multi-document reasoning, standard vector search is insufficient, and legacy graph approaches (GraphRAG/PathRAG) introduce unacceptable cost and latency. While LightRAG presents a highly capable and cost-efficient middle ground, PlantForge successfully tunes the architecture specifically for industrial topology. It provides the deep reasoning of heavy graph models with unprecedented speed and cost-efficiency, making it the optimal architecture for our production deployment.

Why PlantForge Leads

01

Topology-Aware Path Reasoning



Traverses plant topology to find connected information that others cannot see.

02

Verified, Trustworthy Answers



Built-in hallucination verification with citation accuracy that engineers can trust.

03

Performance at Scale



Faster retrieval with optimized paths and flow pruning for real-time industrial use.

04

Built for Industrial Reality



Handles messy, multimodal data, incremental updates, and complex causal relationships.

CONCLUSION

PlantForge addresses a problem routinely misfiled as document management. Thirty-five percent of engineering hours lost to searching, eighteen to twenty-two percent of unplanned downtime traced to fragmented knowledge, and a quarter of India's experienced engineers retiring within the decade are not filing inefficiencies they are a safety, compliance and reliability problem that compounds annually.

Technically, the contribution is treating a plant archive as a topology rather than a corpus. Multi-lane extraction preserves what flat parsing destroys; resolution and denoising make the graph coherent rather than confidently silent; and retrieval extends flow-based path pruning with degree dilution and provenance weighting, so ranking reflects both topological specificity and the reliability of the lane that produced each edge. Every answer is grounded against the evidence actually supplied, and engineer corrections enter the graph as structure letting the system unlearn bad source knowledge without retraining. It is also built to survive deployment: worker pools that scale on backlog, a serialised writer protecting a version guarantee, and measured concurrency work that removed twelve failures per hundred connections.

Commercially, the defensibility is the graph, not the model. Any competitor can call the same LLM; none can replicate a customer's twenty-year maintenance history, its resolved asset identities, or the corrections its own engineers have filed. That asset appreciates with every document ingested and transfers to no one.

What the system deliberately does not do is decide. It drafts, assesses and flags and every output waits for a human signature. In an industry where approval is a legal act, that restraint is not a limitation of the product; it is the reason it can be deployed at all.

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